

Heart Attack Prediction Using Machine Learning. A Data Driven Approach to Identifying Risks.

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Abstract—In the world of advanced medical technologies, early detection of critical conditions like heart disease can significantly improve patient outcomes and reduce healthcare costs. Our research, "Heart Attack Prediction Using Machine Learning: A Data-Driven Approach to Identifying Risks," focuses on developing some predictive models which can evaluate a person's risk of experiencing a heart attack based on his lifestyle and medical data.

I. INTRODUCTION

Using a real world medical dataset, our research applies a full machine learning origin. Where we did data cleaning, exploratory analysis, feature encoding, scaling, and training of multiple classification models. Our goal is to help the healthcare sector in making faster, more consistent, and data driven judgements of heart attack risk.

Our approach not only aims to improve heart attack diagnostic accuracy but also supports preventive care by identifying individuals who are at very high risk. Ultimately, the system demonstrates the potential of machine learning to enhance medical decision making and patient care.

II. DATASET-DESCRIPTION

Total Features: There are 16 features in total. Of them, there are 15 input features (gender, age, education, currentSmoker, cigsPerDay, BPMeds, prevalentStroke, prevalentHyp, diabetes, totChol, sysBP, diaBP, BMI, heartRate, glucose) and 1 output feature (Heart Disease (in next 10 years)).

Total Data Points: 4240

Problem Type: This is a Classification problem. It is a classification problem because the target variable Heart Disease (in the next 10 years) is categorical with output discrete classes (0, 1). The goal is to predict which output class a patient belongs to based on their medical and lifestyle attributes.

Feature Types:

- Quantitative Features (Numerical): There are 14 Numerical features. They are: age, education, currentSmoker, cigsPerDay, BPMeds, prevalentStroke, prevalentHyp, diabetes, totChol, sysBP, diaBP, BMI, heartRate, glucose_loan_percent_income, cb_person_cred_hist_length, credit_score.

- Categorical Features: There is only one categorical feature. It is: gender

Correlation Insights:

- The output feature has a negative correlation with education (-0.05).
- prevalentHyp has strong correlation with sysBP(0.70) & diaBP (0.62).
- Negative correlation is observed with the features age, cigsPerDay(-0.19), currentSmoker (-0.21).
- Strong Inter-Feature Correlations (Multicollinearity) is present between sysBp with diaBp (0.78) and cigsPerDay with currentSmoker (0.76).

Interpretation of Correlation Test: The correlation heatmap suggests that Strong Inter-Feature Correlations are present between sysBp & diaBp and cigsPerDay & currentSmoker. This relation can inflate variance in linear models like logistic regression and make coefficient interpretation unstable. We also saw that feature education has a negative correlation with the output feature. Dropping such negative and strongly correlated features improved our performance.

III. IMBALANCE DATASET

The dataset is imbalanced. The output feature Heart Disease (in next 10 years) has:

- 3596 instances of class 0
- 644 instances of class 1

This means class 0 significantly outweighs class 1, indicating an imbalance that may affect classification model performance unless handled properly as shown in figure 1.

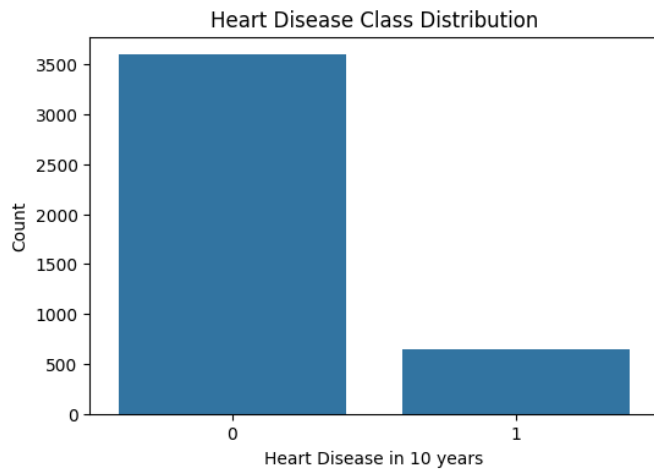


Fig. 1: Heart Disease Class Distribution

B. Distribution of Categorical Feature (Gender):

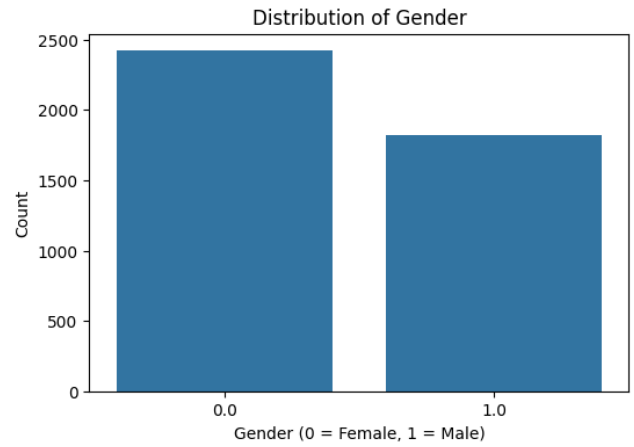


Fig. 3: Distribution of Gender

IV. EXPLORATORY DATA ANALYSIS

A. Distribution of Numerical Features:

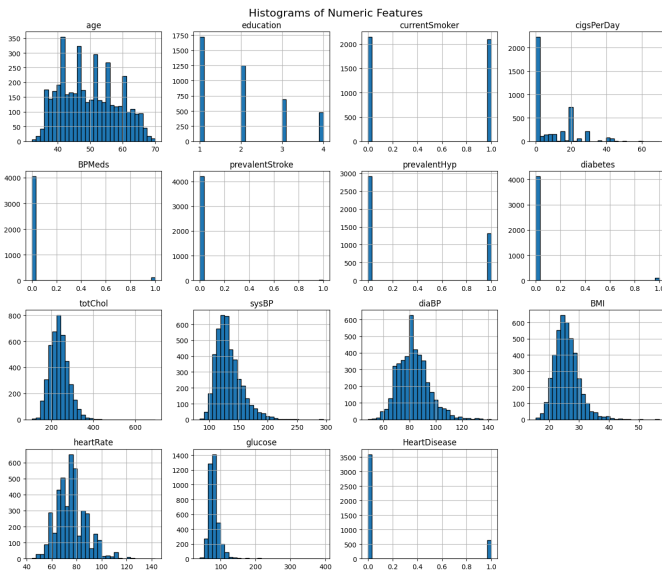


Fig. 2: Histograms of Numeric Features

C. Boxplots to Detect Outliers:

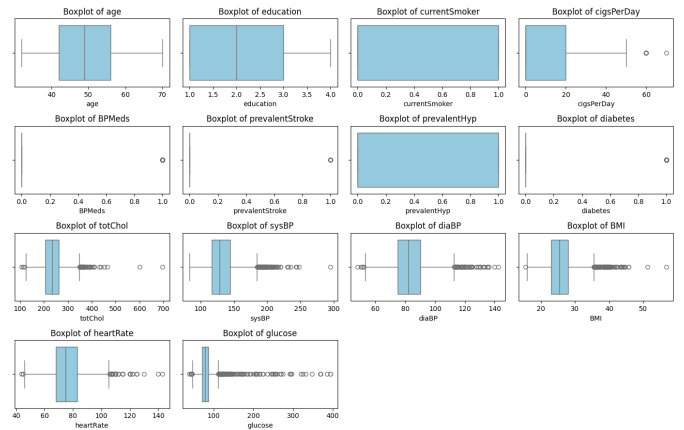


Fig. 4: Boxplots to Detect Outliers

D. Correlation Between Features:

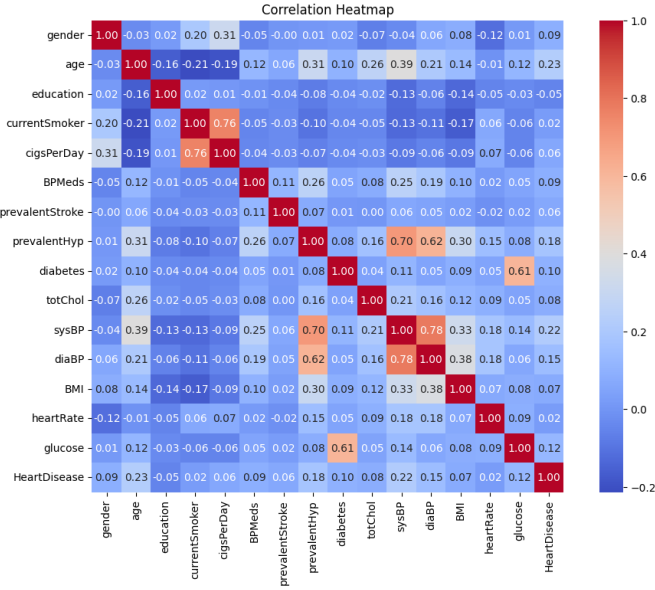


Fig. 5: Correlation Heatmap

V. DATASET PRE-PROCESSING

A. Null / Missing values

We found some null values, but could not find any duplicate rows.

```
Null values in each column:
gender      0
age         0
education   105
currentSmoker 0
cigsPerDay  29
BPmeds      53
prevalentStroke 0
prevalentHyp 0
diabetes     0
totChol     50
sysBP       0
diaBP       0
BMI         19
heartRate   1
glucose     388
HeartDisease 0
dtype: int64

Missing values after imputation:
gender      0
age         0
education    0
currentSmoker 0
cigsPerDay  0
BPmeds      0
prevalentStroke 0
prevalentHyp 0
diabetes     0
totChol     0
sysBP       0
diaBP       0
BMI         0
heartRate    0
glucose     0
HeartDisease 0
dtype: int64

Number of duplicate rows: 0
Shape after dropping duplicates: (4240, 16)
```

Fig. 6: Null values in each column and Missing values after imputation

B. Categorical Encoding

```
gender age education currentSmoker cigsPerDay BPmeds prevalentStroke prevalentHyp diabetes totChol sysBP diaBP BMI heartRate glucose HeartDisease
0 1 39 4.0 0 0 0 0 0 0 195.0 106.0 70.0 26.97 80.0 77.0 0
1 0 46 2.0 0 0 0 0 0 0 0 200.0 121.0 81.0 28.73 95.0 76.0 0
2 1 48 1.0 1 200 0 0 0 0 0 245.0 127.5 80.0 25.34 75.0 70.0 0
3 0 61 3.0 1 300 0 0 0 1 0 225.0 150.0 95.0 28.58 65.0 100.0 1
4 0 46 3.0 1 230 0 0 0 0 0 285.0 130.0 84.0 23.10 85.0 85.0 0
5 0 43 2.0 0 0 0 0 0 1 0 228.0 160.0 110.0 30.30 77.0 99.0 0
6 0 63 1.0 0 0 0 0 0 0 0 205.0 138.0 71.0 33.11 60.0 85.0 1
7 0 45 2.0 1 200 0 0 0 0 0 313.0 100.0 71.0 21.68 79.0 78.0 0
8 1 52 1.0 0 0 0 0 0 1 0 260.0 141.5 89.0 26.36 76.0 79.0 0
9 1 43 1.0 1 300 0 0 0 1 0 225.0 162.0 107.0 23.61 93.0 88.0 0
```

C. Feature Scaling

We used the Minimax scaling technique to scale the data.

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

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Fig. 7: Fig : Feature scaling completed. Here are the first few rows after scaling

VI. DATASET SPLITTING

We split the dataset into 70% for the training and 30% for the testing.

Total Samples	4240 (100 %)
Training Set	2968 (70 %)
Testing Set	1272 (30%)

VII. MODEL TRAINING AND TESTING

In our project, we implemented a supervised machine learning structure to predict the likelihood of heart disease within 10 years. We trained and tested the following models on a cleaned and preprocessed medical dataset:

- **Logistic Regression:** A linear classification model used for binary or multiclass problems. It estimates probabilities using the sigmoid function and predicts the class based on a threshold (usually 0.5). Simple, interpretable, and works well with linearly separable data.
- **Decision Tree:** A tree-based model that splits data into branches based on feature values to make decisions. It handles both classification and regression tasks. Easy to interpret but prone to overfitting unless pruned or ensemble (e.g., Random Forest).
- **Naive Bayes:** A probabilistic classifier based on Bayes' theorem, assuming feature independence. Fast and efficient for high-dimensional data (e.g. text classification). Common variants: Gaussian, Multinomial, and Bernoulli Naive Bayes.
- **Neural Network (MLPClassifier):** A Multi Layer Perceptron (MLP) is a feedforward artificial neural network with hidden layers. It learns complex patterns through backpropagation and activation functions (ReLU, sigmoid). Powerful but requires large data and tuning to avoid overfitting.
- **KNN:** A distance based algorithm that classifies a sample based on the majority class among its 'K' nearest neighbors. KNN is simple and intuitive, but sensitive to feature scaling and may perform poorly on imbalanced and high dimensional datasets. It is particularly useful when decision boundaries are irregular.

VIII. MODEL SELECTION/COMPARISON ANALYSIS

A. Accuracy and Precision, recall comparison of all models (for classification)

No	Model	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.856132	0.700000	0.107692	0.1866
1	KNN	0.838836	0.413793	0.123077	0.1897
2	Naive Bayes	0.829403	0.401786	0.230796	0.2931
3	Neural Network	0.848270	0.555556	0.051282	0.0938
4	Decision Tree	0.757862	0.227053	0.241026	0.2338

B. Bar chart showcasing the prediction accuracy of all models (for classification)

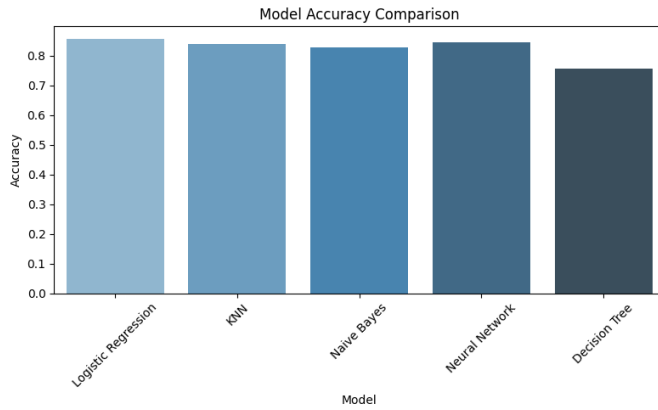


Fig. 8: Model Accuracy Comparison

C. Bar Chart showcasing the comparison of each models F1 Score, Precision and Recall

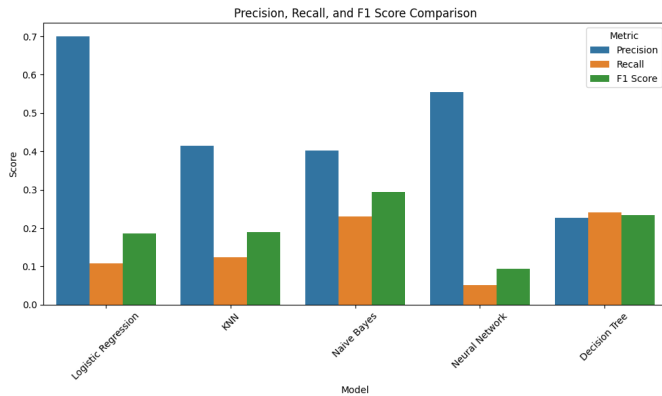


Fig. 9: Precision, Recall, and F1 Score Comparison

D. Confusion Matrix (for classification)

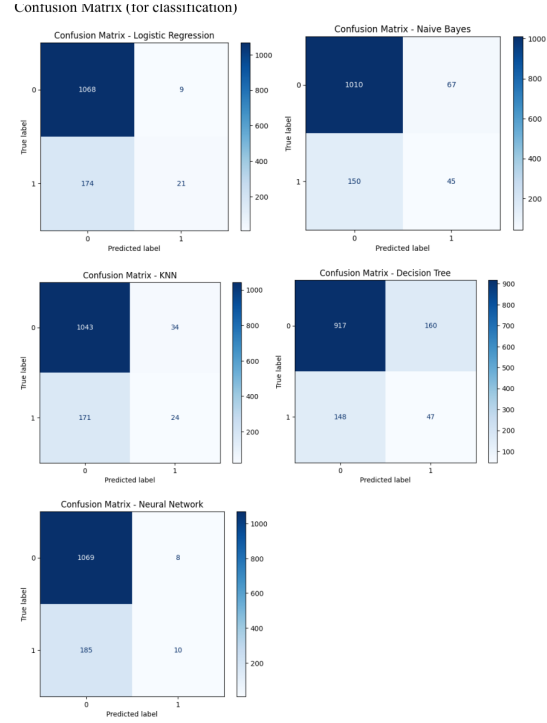


Fig. 10: Confusion Matrix (for classification)

E. AUC score, ROC curve (for classification)

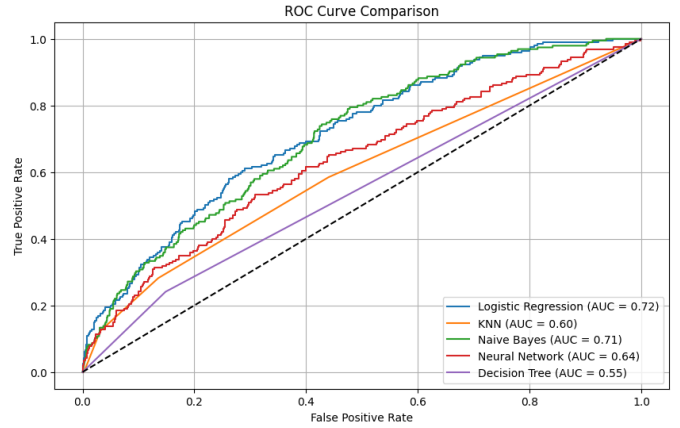


Fig. 11: ROC Curve Comparison

IX. RESULT ANALYSIS

From the evaluated models, Naive Bayes emerged as the most effective in identifying heart disease cases.

- Naive Bayes had the highest recall and highest F1 Score, making it the most suitable model in a medical context, where identifying risky cases is more critical than just overall accuracy.

- Logistic Regression (**85.61%**) and Neural Network (**84.51%**) achieved the highest accuracy, but both models had low recall, meaning they often failed to detect actual positive heart disease cases which is highly risky for medical diagnostics.
- Decision Tree had a lower overall accuracy but maintained a reasonable balance between recall and F1 score, suggesting it can be a viable option after tuning.
- KNN underperformed across all major metrics particularly in recall and F1-score, making it the least effective model in this context.
- AUC Scores indicated that Logistic Regression and Neural Network provided better separation between classes, but again, their low recall limits their utility in high stakes diagnosis.

X. CHALLENGES

- Class Imbalance: The dataset was highly skewed toward "no heart disease" cases, which made learning minority class patterns difficult.
- Low Recall: Most models failed to detect the actual positive cases effectively.
- Overfitting Risk: Especially in complex models like MLP, which require tuning.
- Model Tuning: Especially time consuming for Neural Networks due to multiple hyperparameters.

XI. FUTURE IMPROVEMENTS

To improve performance and reliability, the following enhancements are recommended:

- Address Class Imbalance: Use SMOTE, undersampling, or class weights to help models focus on minority class.
- Feature Engineering: Remove redundant features (highly correlated) and introduce domain-informed features.
- Cross-validation: Implement k-fold cross-validation for more robust and generalized model evaluation.
- Explainable AI: Incorporate SHAP or LIME to explain predictions of black-box models.
- Council with Medical Experts: Validate predictions and insights with medical professionals.

XII. CONCLUSION

Our project successfully demonstrates the application of machine learning in predicting the risk of heart disease using real world medical and lifestyle data. While Logistic Regression and Neural Networks provided high accuracy, they failed to capture positive cases effectively. Despite a lower accuracy, Naive Bayes emerged as the best model in terms of recall and F1-score. Which are crucial in the medical domain where missing high risk patients can be life threatening.

Machine learning has the potential to enhance early diagnosis, which assists doctors in decision making, and enables preventive healthcare interventions. However, the effectiveness of such models depends on balanced data, careful validation, and continuous refinement. With further tuning and integration of medical expertise, such predictive tools could become valuable assets in modern healthcare systems.

XIII. REFERENCE

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